Subject: RE: Q-Ontic Meeting
From: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Date: 5/13/2026, 11:09 AM
To: Lazarovici Dustin <dustin@technion.ac.il>, Pablo Yepes <yepes@rice.edu>

Hi Dustin,

The GREAT merits of Pablo's app are:

1.- There is a continuous transition between a weak (small coupling and large width) measurement and a strong (large coupling and small width) measurement. There are scenarios where the parameters (middle coupling) of the simulation are not clearly in the weak or strong regime, but such scenarios also occur in the laboratory, and we must be able to implement them.

2.- What is done in the Bohmian measurement in Pablo's app is the same for both weak and strong measurements. Basically, the pointer is following the y Bohmian position (as it should). Of course, a measuring apparatus is not just a "y" degree of freedom, but a macroscopic object. However, everybody models the measurement in this way, basically because you cannot simulate a pointer as a full macroscopic object.

>In the weak measurement regime, the coupling is in general too weak for an individual measurement to produce a clearly discernible shift of the >pointer, even if the interaction has taken place. Unless the model implements statistics (where weak measurements become informative through >averages over many runs), one may simply show the pointer as not visibly moving at all. Alternatively, one could show the pointer jittering around i>ts ready position, with measurement-induced shifts that are (in general) small compared to that noise. But this gets more complicated.

Pablo has already introduced the statistics you mentioned (the same for weak and strong measurements), because it is assumed that the y trajectories are in quantum equilibrium in the distribution of the y direction.

>I guess one could implement some model in which the pointer always shifts visibly, while the shift is only weakly correlated with the measured >quantity... But that just doesn't seem to be the setup here.

Perhaps you are arguing that the "observable" being transmitted or reflected only has two outputs, not infinite positions as it has now. What you are suggesting would require an additional step: introducing a line in the detector at the middle position between the transmitted and reflected regions. If the pointer (y position) is above this middle line, the particle is measured as transmitted; if it is below, it is measured as reflected. Then the average number of transmitted particles according to the pointer will be T (the transmission coefficient), and the number of particles detected as reflected will be proportional to the reflection coefficient of the x particles.

In any case, I like as it is now, seeing all the noise, which is in fact not noise but valuable information when taking averages. But, I have no big problem in having only two outputs in the detector (ON =transmitted and OFF=reflected) respecting the statistics of the Y.

>In any case, while there might have been some miscommunication via email, this is about the input I can give and I don't feel the need to discuss >much more via Zoom. I'm happy to step back and let you decide what implementation you want. My only ask is, again, not to show stuff that >makes people believe there is some obvious way to empirically distinguish Bohmian mechanics from standard quantum mechanics.

Now, what the orthodox approach is doing with the pointer is EXACTLY the same as the Bohmian one. THERE ARE NO EMPIRICAL DIFFERENCES.

My point in the previous email was about the word "collapse". The orthodox collapse can be defined as "a wave packet disappearing", which is difficult to discuss. But I suggest defining "collapse" instead as "the pointer is moving", which is much easier to discuss. If you move the pointer is because the Y degree of freedom have been measured (stringly) in the orthodox picture.

What do you think?

All the best,

Xavier Oriols

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Von: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Gesendet: Mittwoch, 13. Mai 2026 11:03
An: Dustin Lazarovici <dustin@technion.ac.il>; Pablo Yepes <yepes@rice.edu>
Betreff: RE: Q-Ontic Meeting

Hi Dustin and Pablo,

I suggest that we (Dustin, Pablo, and Xavier) have an online meeting someday soon to discuss what exactly we want to implement regarding the strong and weak measurements. I think that if we continue only by exchanging emails, we will not clarify things properly, and we may end up driving Pablo crazy.

In particular, the point that worries me about Dustin's argument is the possibility of presenting weak measurements as a measurement that does not move the pointer.

"A cruder strategy (but one that one could also implement in a consistent way) is not to show the pointer moving at all unless there's a clear separation (a couple of \sigma) in y."

A weak measurement always has an outcome! i.e., a pointer moving!! The real issue is whether this outcome is well correlated with the system or not. In a strong measurement, there is a perfect correlation between the measurement outcome and the property of the system. In a weak measurement, however, the correlation can be very poor. For example, the particle may actually be reflected while the pointer indicates transmission.

The merit of the weak measurement is that if you collect all the outcomes of weak measurement (because weak measurements always have outcomes!!!) and make an average with them, you will get a perfect correlation between expectation values of the system and the expectation values of the weak outcomes of the pointer. And, in my opinion, this merit is shown now in the Apps in a wonderful way!!!

Can we arrange an online meeting among us (Dustin, Pablo, and Xavier) to clarify these points in person?
Best,

Xavier Oriols

From: Lazarovici Dustin <dustin@technion.ac.il>
Sent: Wednesday, May 13, 2026 9:22 AM
To: Pablo Yepes <yepes@rice.edu>; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Subject: Re: Q-Ontic Meeting

Dear Pablo,

Thank you, this is evolving really nicely. But the visualization of weak measurements still doesn't look quite right to me.
The main issue (and this is almost certainly related to Xavier's point) is that one sees the pointer moving in the "Orthodox" view, even if there is no collapse.
A secondary issue is that there are instances of measurements described as "weak" where the Bohmian conditional wave function seems collapse but there is no collapse in the orthodox view. This can be attributed to the vagueness of the orthodox view, but my worry is again that it suggests a concrete distinction between the two "theories" that just isn't there under a more sophisticated understanding of standard QM.

One possible solution, as Xavier suggests, is to always apply some collapse of the pointer state. I'm not sure how Xavier would implement that, but for me it's essentially the GRW mechanics: in the pointer coordinate y, the wave function becomes localized on a certain scale (that corresponds roughly to the width of the initial pointer state). In the weak measurement regime, the transmitted and reflected branch have significant overlap in y, so (parts of) both would survive this localization. In a strong measurement regime, only one branch survives.

A cruder strategy (but one that one could also implement in a consistent way) is not to show the pointer moving at all unless there's a clear separation (a couple of \sigma) in y. One aspect that the current visualization doesn't capture is that there is a coarse-graining involved in translating the y-coordinate to microscopically discernible pointer positions. Also from the point of view of Bohmian mechanics, nearby y-configurations would be indiscernible on a macroscopic scale.

In principle, the most complete/sophisticated model would take both aspects into account (coarse-graining and collapse of the pointer state), but I think this gets too complicated.

In any case (but especially if the pointer is always moving), it may be counterproductive to make the markers "R" and "T" on the scale variable as in the current implementation. The point of a weak measurement is that one cannot tell with any certainty (from an individual measurement) whether the particle was transmitted or reflected.

All the best,

Dustin

_____

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Von: Pablo Yepes <yepes@rice.edu>
Gesendet: Dienstag, 12. Mai 2026 18:26
An: Dustin Lazarovici <dustin@technion.ac.il>; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Betreff: Re: Q-Ontic Meeting

Dear Xavier and Dustin,

I've gone through Xavier's points and tried to address each of them. Thank you again for the detailed feedback and for the graphics—they were very helpful.

I've relabeled the collapse option as "Orthodox," since that's the terminology we've been using. Even if we wish it weren't the orthodox or dominant view, that's the landscape we're working within. I've also moved most controls under the expert panel. The default view is now a beginner mode with only a few essential controls, which I hope feels less intimidating for first-time users.

The expected value for the transmitted wave on the detector now updates according to the coupling strength, and I've marked it with a green tick.

I removed the weak/strong measurement buttons. The interaction strength is now determined directly by the coupling and by the width of the pointer wave function (control added). A weak interaction occurs when the T/R pointer wave functions overlap by more than 1%. That cutoff is arbitrary, of course—we can adjust it or expose it as a parameter if that seems useful.

For the Orthodox and MW views, the pointer position on the dial detector is selected randomly within psi^2 of the relevant wave function: the selected branch for strong measurements, or the overlapping region for weak ones. In weak measurements, there is no collapse in the Orthodox view and no world-splitting in MW. For MW with strong measurements, two pointers are shown, corresponding to the different worlds.

I also fixed a couple of bugs: in the previous version, the T/R status for the PW model depended incorrectly on the detector's y-position rather than the actual x-position.

I hope this version illustrates weak measurement behavior a bit more clearly. I'm very open to developing a separate simulation if you think a more focused treatment would help. Suggestions are always welcome.

As ever,

Pablo

Dear Xavier and Dustin,

Xavier, Thank you for your detail answer and the graphics. I have tried to address your various points brought up.

I changed the label of the collapse to "Orthodox", after all that is the language we are using. Even though, we would like that it would not be the orthodox and unique view, that is the reality. I have also hidden most the controls under the expert option. The default view is a beginner view with very few controls. I hope that makes less intimidating for first-comers.

The value expected for the transmitted wave changes on the detector according to the strength of the coupling and it is marked with a green tick.

I have suppressed the Weak/strong measurements buttons. Now, whether the interaction is strong/weak is determined by the strength of the coupling and the width of the pointer wave function. A weak interaction occurs when the pointer T/R wave functions overlap more than 1%. That is an arbitrary cutoff and we could change or introduce a parameter to control it.

For the Orthodox and MW (many-worlds), the position of the pointer on the dial detector is selected randomly within psi^2 of the selected (strong measurement) or overlapping wave functions (weak measurement). For weak measurements there is no collapse for the Orthodox view and no split of worlds in MW. For MW and strong measurements two pointers are shown, corresponding to the different worlds.

I also corrected a couple of bugs, in the previous simulation the T/R status for PW (pilot wave) depended on the y (detector) position rather than the particular position (x).

I hope this illustrates the weak measurement a bit better. But I'll be happy to develop another simulation whether the weak measurement was more clearly illustrated. I am certainly open to suggestions.

As ever,

Pablo

On 5/7/2026 4:42 PM, Lazarovici Dustin wrote:

Dear Xavier,

Thanks, I agree with what you're saying and think this is indeed the best way of understanding weak measurements in standard QM.

I am a bit worried about adding too much complexity to the simulation that might overwhelm or confuse beginners, but if Pablo is willing to take this on, I'd be happy to see the final product.

Best,
Dustin

Sent from Outlook for Android

From: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Sent: Friday, May 8, 2026 12:16:20 AM
To: Pablo Yepes <yepes@rice.edu>; Dustin Lazarovici <dustin@technion.ac.il>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Subject: RE: Q-Ontic Meeting

Dear Pablo and Dustin,

I basically agree with the drawbacks mentioned by Dustin regarding the apps, but I think the solution is not to eliminate the capability to discuss weak and strong measurements, but rather to correct the drawbacks mentioned by Dustin.

Apart from their own intrinsic interest, the ability to discuss weak and strong measurements will also allow us to develop another app for an experiment that, to my knowledge, does not yet exist as an app: the measurement of a weak value. The most famous weak value is obtained by performing a weak measurement of the position and, after a short time, a strong measurement of the position (together with some post-selection). This weak value gives the Bohmian velocity!!! In my opinion, if we are able to make an app representing weak values (a consecutive weak and strong measurement), we will bring the Q-ontic project to a significantly higher level!!!

However, before deciding whether we want to develop an app for weak values or not, we first need to correct the drawbacks in the current weak and strong measurement app mentioned by Dustin. I attach two figures to explain my point of view on what should be represented in the apps. Mainly, what has to be corrected in the app corresponds to points 1, 2, and 3 below:

*** FIGURE 1: *** In figure 1, I plot a unitary evolution on how the system x is coupled to the system y. Still no collapse. But, this unitary evolution has some drawbacks if we want to model the unitary evolution of a weak and a strong measurement.

1.- There are two widths ((\sigma)) in the unitrary evolution: the width of the wave packet in the (x)-direction, which is relatively irrelevant, and the width of the wave packet in the (y)-direction, which is very relevant AND NOW IT CANNOT BE CHANGED. I think it would be better if these two sigmas could be changed and selected independently.

2.- The labels (R) and (T) in the detector should not always be fixed, but should instead indicate the central position of the wave packet in the (y)-direction (for the reflected and transmitted parts). See the plots in figure 1.

*** FIGURE 2: ***
I attach Figure 2, where I explain the "collapse" in weak and strong measurements. To understand what the "collapse" is doing, it is important to notice that a WEAK measurement of the system is, in fact, a STRONG measurement of the ancilla. The ancilla is our Hilbert space (y). Technically, we would need a new degree of freedom (z) to measure the ancilla (y), but we usually ignore it. The STRONG measurement of the ancilla has not yet been performed in figure 1 (only unitary evolution as a pre-measurement). I explain the "collapse" in Figure 2:

3.- In the Bohmian case, the state of the system after the weak measurement of the system (i.e. a strong measurement of the ancilla y) is simply the conditional wave function \Psi(x,Y(t)). In figure 2, we see that some times it is only one wave packet and some times two wave packets. In the orthodox case, we reproduce what has been explained within Bohmian theory, but without trajectories. This, I think, is in agreement with Dustin's previous comments that:
(i) Orthodox theory is not even a complete theory because it does not explain what happens during a measurement, and
(ii) Orthodox results are always equivalent to Bohmian ones.

What do you think? The problem of representing the "measurement" here is because the wave function is indeed \psi(x,y,z) not \psi(x,y). But, we have ignored the process of strongly measuring the ancilla (y degree of freedom)

If anything needs to be explained more clearly, please let me know.

Finally, I want to repeat that this effort to include weak and strong measurements properly opens the door to a truly high-level app: a weak value "measuring" the Bohmian velocity.

Best regards,

Xavier Oriols

From: Pablo Yepes <yepes@rice.edu>
Sent: Thursday, May 7, 2026 7:14 PM
To: Lazarovici Dustin <dustin@technion.ac.il>; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Subject: Re: Q-Ontic Meeting

Dutin,

I have addressed 1 and 2. I hope it makes more sense now.

Xavier,

I am not sure how to handle 3 and 4. What does a weak-measurement mean in this context? In standard QM where is the transition between a weak (no-collapse) and strong (collapse) measurement? Maybe we should just remove the weak and strong buttons. What do you think?

Cheers,

Pablo

On 5/7/2026 6:22 AM, Lazarovici Dustin wrote:

Hi Pablo,

I'm not sure I fully understand what the concept is, but allow me to make a few observations:

1. In the "Projection Postulate" view, the behavior of the pointer is different than in the "Pilot Wave" or "Many-Worlds" view. This might falsely suggest that one could empirically distinguish this "theory" simply by observing how far the pointer moves.
2. The "strength" of the measurement depends not only on the coupling constant but also on the duration of the measurement interaction. Currently, it looks like that interaction persists from the moment the wave function reaches the detector surface to the end of the simulation (and thus depends on the placement of the detector). It might be better to remove that variable by fixing a time window for the interaction between particle and measurement device.
3. With the current default settings for "weak", we are not in what I would consider the weak measurement regime. For me, a "weak measurement" looks more like 5% coupling (in the pilot wave view). That is, from an individual weak measurement, one cannot tell with any certainty whether the particle was transmitted or reflected and the (conditional) wave function wouldn't fully collapse.
4. The same is true in standard quantum mechanics: A single weak measurement (in the technical sense of "weak measurement") wouldn't really collapse the wave function. But I'm not sure exactly what view "Projection Postulate" is supposed to represent, and a weak measurement just isn't a projective measurement to begin with. So perhaps Xavier should say what visualization he had in mind here. For me, it's just important not to give the false impression that one could empirically distinguish standard QM from BM by showing differences that don't really exist under a somewhat sophisticated understanding of standard QM.

Best,
Dustin

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https://protect.checkpoint.com/v2/r02/___www.technion.ac.il___.YzJlOnRlY2huaW9uOmM6bzpmODE3ZWVlOGVhYWMwNjMzYzgyY2U2ODgyY2UxNjQwMTo3OjE2MWQ6MGU0NjM2ODM5ZDc1YzE3MjE5ZDhlZGVkYTFjMGM5ODZhNThlNTgyM2E2MzVlYWUwOTY3OTI2MjE2YzFmYmE1YTp0OlQ6Rg
\ dustinlazarovici.com

Von: Pablo Yepes <yepes@rice.edu>
Gesendet: Mittwoch, 6. Mai 2026 05:13
An: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; Dustin Lazarovici <dustin@technion.ac.il>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Betreff: Re: Q-Ontic Meeting

Xavier,

Check the new version, I think it has all you suggested: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/measurement___.YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OjQzN2Q6MjA2MTQxZDBlMzZiYzgxNTZhMmU2NGZmN2EzZWQ4ZjlkMjVlMWNlZDE5OGM0ZjI2YTFjODk5ZmQwZjM0NTAzODp0OlQ6Rg. I do think that the more we work on it, the better the apps will be. This is an iterative process...

I also implemented your 2D simulations from the Matlab code. Have a look at https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/dev___.YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OjY1MzA6MGNlMjAyZGY2MDVmMTFmMmViYjRjNWJmY2YyMTZiOTllMGU0MDUxZmVkY2VjMjhjMWQ0OTc0Y2VmNjVhYjU0MTp0OlQ6Rg.

All,

I added a page with everyone in the collaboration: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/collaborators.html___.YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OjVhNDg6ZWY1YTY1MzNkNmNmZjgxMjA1Yzg1OTgzYjAxNGNjZDVjOGI4MWNmMTBiOTk3NjRhZGY0NjVmODlkZDRjOTI4ODp0OlQ6Rg. Please let me know if you are ok with it. Right, it can be accessed from the bottom of the main page. We are listed in alphabetic order. Some pictures may need an update...

Anssi,

I'll be happy to add you if you would like to be included.

Cheers,

Pablo

On 5/4/2026 5:28 PM, Xavier Oriols Pladevall wrote:

Hi Pablo,

I like very much how the setup looks now.

Since you are always asking for suggestions—and you have not complained (yet) that writing suggestions is much easier than finally implementing them in the app —I will add one more for you to decide whether it makes sense.

It is about the role of the coupling parameter that, together with the wave-packet width, allows one to effectively simulate both strong and weak measurements:

1. Strong measurement: A strong measurement corresponds to a small pointer wave-packet width together with a large coupling. In this design, there is a clear separation between the pointer states associated with the transmitted and reflected components.

2. Weak measurement: A weak measurement corresponds to a small coupling and a large pointer wave-packet width. In this regime, there is a significant overlap between the pointer wave packets corresponding to transmitted and reflected components, so one cannot determine with certainty whether the particle was transmitted or reflected.

I think this ability of the app to reproduce the design of both strong and weak measurement apparatuses is a very nice feature. If we decide that this weak and strong visualization is worthwhile, then I believe the following points should be modified:

a. The movement of the detector arrow (i.e., its angle of rotation during the measurement) should be directly proportional to the pointer particle's motion in the vertical direction. In other words, a small vertical displacement of the pointer particle should correspond to a small change in the detector angle.

b. The difference between the angles defined as "transmitted" and "reflected" in the detector should be proportional to the separation between the central positions of the pointer wave packets corresponding to transmitted and reflected events in the vertical direction.

With these modifications, users of the app will understand that sometimes a measurement apparatus is not sufficiently accurate—for example, due to wave-packet pointer overlap, a weak measurement may lead to incorrect identification of the Bohmian particle's behavior. This will not happen in a strong measurement, where the correlation between the particle's behavior and the detector's indication is perfect.

In fact, with the proposed modification of the pointer's angular response, we recover what Dustin is expecting: a small coupling leads to a small displacement in the vertical direction and, consequently, a small movement of the pointer angle.

Sorry for the long email.

All the best,

Xavier Oriols
P.S. Regarding collapse, I was initially thinking about continuous collapse—i.e., a sequence of POVM-type collapses occurring at different times during the visualization, not only at one time (where the POVM collapse does not eliminate one of the wave packets, but instead gradually increases the probability of one of them and reduce the other)—but I fully agree that it is better to forget about it

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From: Pablo Yepes <yepes@rice.edu>
Sent: Monday, May 4, 2026 10:08 PM
To: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; Lazarovici Dustin <dustin@technion.ac.il>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu; Anssi Kuha <Anssi.Kuha@metropolia.fi>
Subject: Re: Q-Ontic Meeting

Xavier,

I've implemented your suggestions 1 and 2.

On point 3, I've labeled it "Projection Postulate" and added a tooltip explaining the meaning. I'm hesitant to call it "Orthodox" or "Textbook" QM, since part of the project's aim is precisely to unsettle that framing. As you showed in an earlier draft, I also experimented with a 1-D representation of the measurement process for the Copenhagen view, but the results weren't very satisfying. If you have a more concrete suggestion—or a Matlab implementation—I'd be glad to try it.

Dustin,

Regarding your suggestion to include a spontaneous-collapse option: I worry it may introduce more confusion than clarity. Although it is conceptually distinct from Copenhagen, the visual behavior would be nearly identical in this interface.

Further comments and suggestions are very welcome.

Pablo

On 5/1/2026 1:25 PM, Xavier Oriols Pladevall wrote:

Dear All,

I like the app for measurement that Pablo developed. Just a few ideas that perhaps (not sure) could be useful:

1. I think it could be enlightening if, in the app, it were possible to choose (i) whether the barrier is present or not, and (ii) whether the measuring apparatus is present or not, and then run the simulations accordingly. See the attached videos as examples. It makes clear that the act of measurement is not just "seeing" the particles, you have to include a new apparatus!
2. It would be nice if the degree of freedom of the pointer (vertical direction on the right) could be represented as a moving arrow of a measuring apparatus. In addition to the (conditional) wave packet in the vertical axes, it would be useful to also show an arrow representing the measuring device. If the arrow does not move, you detect nothing!
3. I think the name "Copenhagen/Collapse" is still ambiguous. Spontaneous collapse theories (GRW) are not "Copenhagen/Collapse" and they provide a well-defined mechanism for collapse. Perhaps (not sure) we could refer to what is now named "Copenhagen/Collapse" as "standard University interpretation" or something similar, to make it clear that what we are really is what is taught in (almost) all universities. Spontaneous collapse could also be added as a separate option.

On the other hand, I have developed a MATLAB code for the app that Pablo has created. It is not polished, but I've attached it too if needed. I also attach three videos created with the matlab code:

1. *wavepacket_evolution_barrier.gif*: where you see 1D tunneling in a 2D scenario.
2. *wavepacket_evolution_barrier_measurement.gif*: where you see 1D tunneling including a detector of charge after the barrier.
3. *wavepacket_evolution_measurement.gif*: a detector without a barrier. You will see that some (very few) particles are not detected due to the presence of the detector: the detector makes the particle to be reflected! (I think some Bohmian researchers may find this result interesting, while others may not. See why in : https://protect.checkpoint.com/v2/r02/___https://link.springer.com/article/10.1007/s10701-024-00798-y___YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OmQyOWI6NjNkYzU4NGQ3ZTc2YmM5MzA0OGFlYjBlOWViZTM0Y2NmYmFkOWNjYTUyZmJjODE5NDM4ZjI5NjA1ODU5YWY2MDp0OlQ6Rg).

All the best,

Xavier Oriols

From: Pablo Yepes <yepes@rice.edu>
Sent: Tuesday, April 28, 2026 10:11 PM
To: Lazarovici Dustin <dustin@technion.ac.il>; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu
Subject: Re: Q-Ontic Meeting

Dear all,

I have updated the measurement app: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/measurement___YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OmEwY2Y6ZTY2NjE0OGM1Mjk3M2E4ODQyNzE2NjEzOWVkNjkwMmEwMTNiYThjOGZlNTM1ZDEwODg0YTFmOTVkODIwZjIyODp0OlQ6Rg. The changes are:

1) Introduce a detector after the barrier

2) Eliminated the 1d (operator) simulation.

3) Change the name of Copenhagen view to "Copenhagen/Collapse". I hope that works for everyone. The collapse now happens when the particle hits the detector.

Please let me know what you think.

Thank you,

Pablo

On 4/23/2026 4:44 AM, Lazarovici Dustin wrote:

Dear Pablo,

Great, this looks much better and it also displays pretty well on my phone!

I'm not sure what it means to simulate an "operator view" and whether that simulation is instructive.

But one suggestion: Why don't we turn what is now "Copenhagen + Apparatus" into a simulation of spontaneous collapse? It would look essentially the same -only that the collapse should set in sooner, I think (but perhaps it should do so anyway).

Then the "Copenhagen" or "orthodox" view can be whatever Xavier says it is (Bohr would probably say that one shouldn't try to visualize the wave function in the first place), and the view of standard QM that makes sense to me would essentially be illustrated by "spontaneous collapse".

Best,
Dustin

Dr. Dustin Lazarovici
Assistant Professor of Philosophy
Humanities and Arts Department

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https://protect.checkpoint.com/v2/r02/___www.technion.ac.il___.YzJlOnRlY2huaW9uOmM6bzpmMGJkY2MzMmY0ZDc3MDkzYjM5YjY3YTJlMzY2ZDU3ZDo3OjM3NDk6NjEzYjgzMTJlMTAyNWE0NzkzZDI5YTZkZjA1NWZkM2YwNDVmNjhjZDA4NWIyMTQzZDMxMDVlNGNjNDM3YzdkMzp0OlQ6Rg
\ dustinlazarovici.com

From: Pablo Yepes <yepes@rice.edu>
Sent: Wednesday, April 22, 2026 23:49
To: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; Dustin Lazarovici <dustin@technion.ac.il>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>
Subject: Re: Q-Ontic Meeting

Dustin, Xavier,

Thank you for your comments and feedback.

Dustin,

I think I have address all your four points. I also try to make it responsive, and I was successful to some degree. You can see it on your phone in horizontal mode, however in portrait, it still cuts the canvas. I have also added a QM operator version of the measurement. I also added a many-world option. I hope it has not become too complicated. Please have a look at the new version online: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/measurement___YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjNmNmI6NTRhNmE2ODZmYzA1YzMyODQ0NTBlNWIxNTNmMDY0ODBmOTliZmUxOWMyNWY2Njg4ZjRlYWNhNDkxZjQ1ZDc0YTp0OlQ6Rg.

Xavier,

Concerning modeling the measurement for a non-spatial variable, as you said, we already have the simulation of Stern-Gerlarch, which seems to be close. I would have to think specifically how to implement the measurement process explicitly. Certainly MATLAB programs to implement more realistic simulations of the measurement problem would be very helpful.

As ever,

Pablo

PS: Concernig the ontology of the 3D (condidional) wave function, a parallel argument applies to the nomological reading of Bohmian mechanics advanced by Dürr, Goldstein, and Zanghì. In that framework, the only ontological primitives are particle positions, and the universal wave function Ψ plays a nomological role — it governs particle motion without itself being a material field. Yet particle trajectories depend on Ψ just as fully as the 3D wave functions do: without Ψ, the guidance equation has no content. One might object that particle positions are more primitive because they are simpler — just three numbers rather than a field. But this is a difference in mathematical complexity, not a difference in kind. A particle position can itself be understood as extracted from the full configuration by fixing all other coordinates, just as the 3D wave function is obtained by conditioning Ψ on the actual positions of the remaining particles. In both cases, the mathematical object used to define or govern the ontological primitive is not thereby forced into the ontology. The dependency objection therefore carries no more force against the idea of granting ontology to the 3D wave functions than it does against the particle-only ontology of DGZ.

On 4/20/2026 4:41 AM, Xavier Oriols Pladevall wrote:

Hi Pablo,

I like the applet very much. Apart from Dustin's comments, I would like to add the following:

1. In the orthodox Copenhagen picture, measurements are typically described in 1D, not in 2D. This means that, in most QM textbooks, the measurement is not modeled by a physical measuring apparatus, but rather by an operator (a projector, or a POVM). Therefore, I think it would be nice if this operator-based picture of measurement could also be visualized in a purely 1D setting, which is what students usually learn. (This comment is linked to the idea of "naïve realism about operators.")

2. In the app, the measurement essentially corresponds to detecting transmission or reflection, which ultimately amounts to measuring position. In other words, from the very beginning, one is measuring a quantity (the position) that is

already well defined in Bohmian mechanics. However, this is not always the case. For example, one may measure the energy of a system prepared in a superposition of different energy eigenstates, or measure spin of a system prepared as a sum of spin up and down. In such cases, the value of the "measured" energy or spin is not a pre-existing value, but rather one that is created by the measurement process itself. We could also include this type of measurement. I think the Stern–Gerlach app is very close to what I say here.

3. If needed, I could try to provide a MATLAB program, similar to the ones we have developed before for the double-slit experiment, to implement more realistic simulations of the measurement process. Since 2D simulations are comparable in complexity to those we have already developed, it seems feasible that such a program could run efficiently.

All the best,

Xavier Oriols

P.S.

"It looks to me that in the end, BM explains the measurement problem and effective collapse of the wave function when looking at the conditional wave functions. But isn't that somehow giving a more prominent ontological role to the conditional wave functions than to the configuration space wave function?"

The 1D conditional wave function requires the 2D configuration-space wave function in order to be defined, whereas the 2D configuration-space wave function does not require the 1D one. This seems to be an argument against your suggestion, although I am not sure how strong my argument above really is. (Bohmian trajectories also require the full configuration-space wave function.)

From: Pablo Yepes <yepes@rice.edu>
Sent: Friday, April 17, 2026 4:35 PM
To: Lazarovici Dustin <dustin@technion.ac.il>; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu
Subject: Re: Q-Ontic Meeting

Dear all,

Have a look at https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/measurement/___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OmNmOWY6ZWQ1ZGM0OTI0MGE5NDFhMzQ2MDM2ZGVkNTIzMzU4M2M0YmViZDQ5OGZhZjVlZjUzNmUyZjkYjVjNGFhNTVhNDp0OlQ6Rg. It tries to replicate Xavier's PowerPoint. I am not fully satisfied by how the wave is split between reflected and transmitted, though. Please let me know whether this gets the idea and what needs to be improved.

It looks to me that in the end, BM explains the measurement problem and effective collapse of the wave function when looking at the conditional wave functions. But isn't that somehow giving a more prominent ontological role to the conditional wave functions than to the configuration space wave function.

As ever,

Pablo

On 4/16/2026 7:24 AM, Lazarovici Dustin wrote:

Hi Xavier,

Okay, let's bracket was has been said or not said. I still don't think it is a good idea to include collapse dynamics in the double-slit simulation.

First of all, to really simulate the Bohmian collapse, one would need a full quantum-mechanical model of the screen. This is extremely difficult. In fact, no one really knows how to do it in a somewhat realistic manner; it's an open research problem that I and (from a somewhat different perspective) Larry are working on.

One could put in some collapse by hand — maybe we'll have to do that anyhow to illustrate spontaneous collapse theories. But I don't see much pedagogical value in showing that, if one looks very closely, the collapse in BM looks a little more gradual than in orthodox QM. I don't think a user would understand what this means, why it matters, or what problem it's supposed to solve. We need dedicated applications showing the measurement process on configuration space to explain Bohmian concepts like conditional wave functions and effective collapse.

Finally, it's not clear if illustrating such a difference between BM and orthodox QM is even correct. As you said yourself, standard QM doesn't really make competing predictions about the collapse dynamics - it's just notoriously vague about it. But if one cares about time-scales, a sophisticated "orthodox" physicists would probably try estimating/modeling decoherence times, and then it looks again pretty much the same as in Bohmian mechanics.

The existing simulations don't include an "orthodox view" at all, and I think that's for good reason. When it comes to the details, it's unclear what this "view" is actually supposed to look like in a simulation. From a Bohmian perspective, I would say that "orthodox QM" or "standard QM" is not a competing theory (in fact, it's no serious theory at all) but just an FAPP description of the measurement formalism that can be derived from, and made precise within, Bohmian mechanics. From this perspective, trying to show things that look differently in BM and QM is also off-target because QM really *is* BM.

Best,

Dustin

_____

Dr. Dustin Lazarovici
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Humanities and Arts Department

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\ dustinlazarovici.com

From: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Sent: Wednesday, April 15, 2026 12:37
To: Dustin Lazarovici <dustin@technion.ac.il>; Pablo Yepes <yepes@rice.edu>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>
Subject: RE: Q-Ontic Meeting

Hi Dustin,

Thanks for the email. First of all, I apologize if any of my comments made you feel uncomfortable or if it seemed like "I objected to a bunch of stuff you didn't say." The main motivation of my E.mails was to discuss pedagogical aspects of the apps we are developing.

In this regard, I think your sentence — "If the simulation only shows the particle wave function, it would look FAPP the same in BM and in standard QM" — opens a quite relevant pedagogical discussion on how to treat collapse in the applets. I elaborate on this:

1.- In my own words, your sentence means that, in the double-slit experiment, the collapse occurs on such a fast timescale that "it would look FAPP the same in BM and in standard QM." I fully agree that this is true for all practical purposes (FAPP), but I am arguing that the statement is not necessarily true for fundamental purposes. If we try to look at the timescales at which collapse occurs in BM and in standard QM, they would not look "the same," because the time-evolution process of collapse is undefined in standard QM.

2.- My main pedagogical interest is to present Bohmian theory as clearly and effectively as possible, which for me (although I understand this second point may not be shared by everyone) also involves highlighting the important drawbacks of standard QM. The most important example of this drawback is the wave function collapse in orthodox QM.

I would be glad if the collapse problem of standard QM could be visually manifested in all apps. Against this preference, I recall the comment by Lawrence Frolov that trying to show too many things may end up making the apps impossible to understand.

• Is it possible to visually manifest, within the double-slit experiment, the problems of the orthodox collapse? For example, by introducing a time "zoom" to compare Bohmian and orthodox collapses? The conditional wave function in the Bohmian case and ¿? in the orthodox case because the orthodox theory has not conditional wave function.

• Would it be better to illustrate the issue of standard QM collapse in other experiments where the measurement process (branching) is not so fast on the simulation timescale, such as the Stern–Gerlach experiment?

Again, I apologize if any my comments made you feel uncomfortable. When you have time, I would be happy to hear your opinion on the visualization of BM and SM collapse in the apps.

All the best,

Xavier Oriols

From: Lazarovici Dustin <dustin@technion.ac.il>
Sent: Tuesday, April 14, 2026 10:15 AM
To: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; Pablo Yepes <yepes@rice.edu>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu
Subject: Re: Q-Ontic Meeting

Hi Xavier,

I don't see how this is responding at all to what I wrote. I was reacting to your proposal to include the orthodox collapse in the existing simulation of the double-slit experiment, in order to illustrate how "the wave function, which occupies all space during many time of the simulation, is "suddenly" transformed into a wave function localized only around the position where the particle is detected on the screen." I was pointing out that — in this existing simulation, which doesn't show the wave function on configuration space — the Bohmian collapse would look essentially the same.

Then you object to a bunch of stuff I didn't say.

I agreed with the idea of creating other apps to show how BM handles measurement and collapse; in fact, I had suggested that from the very beginning, hence the slides.

Best,

Dustin

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From: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Sent: Monday, April 13, 2026 16:01
To: Dustin Lazarovici <dustin@technion.ac.il>; Pablo Yepes <yepes@rice.edu>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>
Subject: RE: Q-Ontic Meeting

Hi Dustin,

Thanks for your comments. I would like to respond to them:

>But regarding your second point - showing the "miracle of collapse" in the existing simulations - the issue is that in Bohmian mechanics, the wave function of the particle, strictly speaking, collapses as well. Only >that it's not a miracle; it follows in >a precise way from the definition of conditional wave functions. But to show that, one really needs to look at the wave function of particle + detector on configuration space.

Yes, we fully agree. If you look at the third slide of my PowerPoint, you will see the wave function of the particle + detector in configuration space, as well as two (conditional) wave functions: one for the reflected particle and another for the transmitted one.

>If the simulation only shows the particle wave function, it would look FAPP the same in BM and in standard QM.

I think your sentence is somewhat misleading because it compares apples and oranges. They may look similar for all practical purposes (FAPP), but if one looks at the details, they are totally different:

1. **"Oranges" (standard QM):** there is no explanation of the detailed mechanism of collapse of the wave function (when it occurs, over what timescale, or what kind of wave function change is involved). The theory simply does not address these details.

1. **"Apples" (Bohmian mechanics):** on the contrary, in Bohmian mechanics all details of the effective collapse are fully determined and, in principle, predictable (when it happens, over what timescale, and the precise form of the resulting wave function).

I find it unfair to say that BM and standard QM describe the same "particle wave function" evolution during the measurement because standard quantum mechanics does not describe the collapse, just postulates how it ends. In fact, strictly speaking, your statement is not only unfair but also, in my view, incorrect. The "effectively collapsed" (conditional) Bohmian wave function satisfies a non-unitary Schrödinger-like equation (the equation of motion of the conditional wave function), whereas there is no corresponding non-unitary Schrödinger equation for the particle wave function in standard quantum mechanics. So how can the two wave functions be the same if their equations of motion are different? That does not seem possible.

I am quite sure we also agree on this, and it would be nice if a "collapse as a miracle occurs" could be somehow represented in the apps.

All the best,

Xavier Oriols

From: Lazarovici Dustin <dustin@technion.ac.il>
Sent: Monday, April 13, 2026 1:56 PM
To: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; Pablo Yepes <yepes@rice.edu>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu
Subject: Re: Q-Ontic Meeting

Hi Xavier,

I very much agree with the goal of showing that BM is about more than adding trajectories.

I resend the slides I shared a while back, which also contain some animations that could be turned into a simulation. There is also this nice video that Stefan Teufel created many years ago:
https://protect.checkpoint.com/v2/r02/___https://www.youtube.com/watch?v=fNFbkLMljMc&t=308s___YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjA5NTI6YjM3ZDU5YzBjZjk0MjFmMjJlMTZkODdjN2Y2ODdkMDA3ODBhYjUzNDY3MjI1YzQzOTY3NWUzOGU1NGUxYWZjNzp0OlQ6Rg

But regarding your second point - showing the "miracle of collapse" in the existing simulations - the issue is that in Bohmian mechanics, the wave function of the particle, strictly speaking, collapses as well. Only that it's not a miracle; it follows in a precise way from the definition of conditional wave functions. But to show that, one really needs to look at the wave function of particle + detector on configuration space. If the simulation only shows the particle wave function, it would look FAPP the same in BM and in standard QM.

Best,

Dustin

_____

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\ dustinlazarovici.com

Von: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Gesendet: Montag, 13. April 2026 14:21
An: Pablo Yepes <yepes@rice.edu>; Dustin Lazarovici <dustin@technion.ac.il>; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be <jean.bricmont@uclouvain.be>; js84@rice.edu <js84@rice.edu>; laf230@rutgers.edu <laf230@rutgers.edu>; shadit@math.rutgers.edu <shadit@math.rutgers.edu>
Betreff: RE: Q-Ontic Meeting

Dear all,

In our last meeting, we discuss that it would be a pity if all the effort we are putting into developing the apps ended up conveying the message that Bohmian mechanics is simply about adding trajectories on top of the wave function to reproduce the same evolution already described by the wave function (with different colors). This is, to some extent, how Bohmian mechanics is often presented, but it is not a very compelling way to convince people about its merits. We also discussed that one of the best ways to appreciate the merits of Bohmian theory is through how it explains measurement and clarifies some of the "miracles" required in the orthodox formulation. Regarding this point, I have two comments:

Comment 1:
I attach three slides from PowerPoint material that are intended only as inspiration for developing an application based on these ideas. The example involves the measurement of a particle that is either transmitted or reflected. In the Copenhagen picture, the wave function splits into transmitted and reflected parts, and at some point a "miracle" occurs in which the particle is found in one outcome or the other, but not both. In contrast, in the Bohmian framework, no such miracle is needed: one simply solves the Schrödinger equation in 2D, including the measuring apparatus, and the different outcome follows naturally in the 1D effective space. To my knowledge, this is something that is largely absent from public explanations and YouTube content on Bohmian mechanics, so it could be quite impactful. We do not need to focus on this particular example, it just wants to be an inspiration/motivation.

Comment 2:
In the apps that Pablo has already developed for the two-slit experiment, it would be wonderful if we could also illustrate the orthodox "miracle (i.e. collapse)" in the simulation. By this, I mean that the wave function, which occupies all space during many time of the simulation, is "suddenly" transformed into a wave function localized only around the position where the particle is detected on the screen. The simulation will look quite bizarre, with the wave function randomly collapsing from everywhere into a point, but this is not our fault; it is simply how orthodox quantum mechanics is formulated.

All the best,

Xavier Oriols

P.S. Later, I will send the improved versions of the word documents on the special issue we have discussed.

From: Pablo Yepes <yepes@rice.edu>
Sent: Tuesday, March 31, 2026 5:51 PM
To: Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>; dustin@technion.ac.il; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu
Subject: Re: Q-Ontic Meeting

Dear Xavier,

I think the email and rationale are in good shape. Here are a few suggestions that might strengthen the case.

In the rationale, the opening mentions the second quantum revolution but then drops it. There is a powerful argument there that is worth spelling out: students being trained for quantum computing, cryptography, and sensing arguably need a deeper ontological grasp of what they are engineering, not just calculational fluency. Making that connection explicit would sharpen the motivation considerably.

The paragraph on "shut up and calculate" is your strongest moment. You could reinforce it by noting that physics education research has consistently shown that conceptual understanding — not just problem-solving — is what distinguishes expert from novice physicists. That gives the critique an empirical grounding rather than a purely philosophical one.

You should also preempt the obvious objection: a skeptical editor will immediately wonder whether exposing students to contested interpretations will just confuse them. A sentence or two clarifying that the goal is pluralism and critical thinking — not replacing one orthodoxy with another — would disarm that before it arises.

On Bell: the phrase "largely unsuccessfully" undersells him. Something like "nearly 50 years ago, John Bell had the foresight to raise this question — a debate that, despite his efforts, never fully took hold in mainstream curricula" reframes his role as visionary rather than simply unsuccessful, and makes the special issue feel like a long-overdue vindication rather than a revival of a failed project.

In the email itself, the offer to guest-edit is a bit hedged. Stating clearly that we are proposing to serve as guest editors, and briefly noting why the group is well positioned — the breadth across applied, fundamental, and educational expertise is a genuine strength — would read with more confidence.

I wonder if it would help to indicate that we think a good number of authors would contribute to the special issue.

Best,

Pablo

On 3/31/2026 7:22 AM, Xavier Oriols Pladevall wrote:

> Dear all,
>
> Yes, there is another topic that we still have ongoing and not yet completed, which is the special issue we discussed.
>
> I have looked into the best way to submit the proposal. In reviewing several of the consulting editors listed for the *American Journal of Physics*, it is clear that there are few of them who are highly knowledgeable in quantum mechanics—who have even written books on quantum mechanics and quantum field theory. For example, the three listed below
>
> **David J. Griffiths** (AJP Consulting Editor)
>
> Well-known textbook author (*Introduction to Quantum Mechanics and Introduction to Quantum Electrodynamics*) and long-time contributor to AJP.
>
> https://protect.checkpoint.com/v2/r02/___https://www.reed.edu/physics/faculty/griffiths/___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjJiZTg6OTRmMDQzMmQxMGZmYTE0NWVjMTk3YmEwN2I5NzdkMGJhMGUyMjIzNzk4Njk1Y2EwZGJlNzYwYjI3NjZlMDcwYTp0OlQ6Rg
>
> **Daniel V. Schroeder** (AJP Consulting Editor)
>
> Co-author of *Quantum Field Theory* textbooks and a respected physics educator.
>
> https://protect.checkpoint.com/v2/r02/___https://physics.weber.edu/schroeder/___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjU5MGI6NDMwNDQxYzViMmI3ZWM5NjY3MmQ5Mjk5MjM5MWI4MmFmYjA2MWFhNWIwMjM4ZmFhNDA4YjYyYzUzZTYzZWM1ZTp0OlQ6Rg
>
> **John L. Bohn** (AJP Consulting Editor)
>
> https://protect.checkpoint.com/v2/r02/___https://www.colorado.edu/physics/john-bohn___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjVjZWE6MzY1NjE5YjAzMGI3NjQzM2E2NmJiYTNhZTUzMGRmMDk5MTE5NjVkMjQ5ZTI3MzQzMGMwNjFlODE3MDQ0ZDAzMDp0OlQ6Rg
>
> However, I am not sure that they would be very receptive to these realistic (hidden-variables) interpretations of quantum mechanics. Perhaps yes, but not sure. Another possibility is written directly to the editorial office and wait for their response.

AJP Editorial Office:
Professor Beth Parks
Department of Physics and Astronomy
Colgate University
Hamilton, NY 13346
ajpeo@aip.org

If we finally decide to E,mail the editorial office, I am including here a draft of the email, along with the previous text we had already discussed regarding the special issue. I have also updated the title to make it more engaging.

I would appreciate it if we could discuss this issue this afternoon to decide how finally proceed.

All the best,

Xavier Oriols

From: Pablo Yepes <yepes@rice.edu>
Sent: Monday, March 30, 2026 11:41 PM
To: dustin@technion.ac.il; e.tang@rice.edu <et50@rice.edu>; jean.bricmont@uclouvain.be; js84@rice.edu; laf230@rutgers.edu; shadit@math.rutgers.edu; Xavier Oriols Pladevall <Xavier.Oriols@uab.cat>
Subject: Re: Q-Ontic Meeting

You can see a development website at: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/dev___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OmM1N2E6NmJlMjEzNjMwOGNkOTUxMTc3MDQ3ZDFjZTY5YjE1ZTlkYTliMTE1MjQ1OWNjYjUxNGNjNTQ0YzQ3OTA4NGJlZTp0OlQ6Rg .

It has an additional app for the free particle and at the end a link to a page with the collaborators.

Talk to you tomorrow,

Pablo

On 3/30/2026 9:01 AM, Pablo Yepes wrote:

> Dear all,
>
> This is a friendly reminder that we are meeting tomorrow, March 31, at 8 AM CDT.
>
> Please let me know if there is any topic you would like to bring up.
>
> I can give you an update on where we are:
>
> 1) Website: Please take a look at https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OmMyNzM6ZDJhNzA4OTI2ODNlMzk4YzJiYzExMTgyNzAxYmFlNThmMTc4YzdlYTY2OWRlMDZiNDY2NWI1MjY3MzMwZDk1Mzp0OlQ6Rg — we've added several new simulations. We are still working on one on Bells correlations and one on tunneling. I'm leaning toward rebuilding each simulation from scratch with AI assistance, which seems easier and more flexible than continuing to develop the universal QFT framework we started, but the decision isn't final yet. We also have a list of simulations we want to implement here. We should probably start implementing them systematically.
>
> I'd also like to add a collaborators page to the Qontic site. Please tell me if you're happy to be included.
>
> 2) We have interviewed four postdoc candidates, and have one interview pending; two candidates look very strong. We are also hiring a undergrad to contribute to the project over the summer. We have received the funding from FEHE (supportive foundation).
>
> 3) Jared and I would appreciate your experience and advice on teaching pilot-wave theory to undergrads. We are considering a course on the various ontologies of quantum mechanics, including pilot-wave theory.
>
> 4) We did not submit a proposal for a session in the PERC 2026, but most probably will submit an abstract.
>
> As ever,
>
> Pablo
>
> On 2/23/2026 1:16 PM, Pablo Yepes wrote:
>
> > Reminder, we have a meeting tomorrow at 8 AM CST. Proposed agenda: 1) Special Issue: Dustin, Xavier; 2) Numerical waves: Xavier ; 3) Update on qsf-v1 and qsf-v2. You can see version 2 at: https://protect.checkpoint.com/v2/r02/___https://qonticlab.rice.edu/qsfv2___.YzJlOnRlY2huaW9uOmM6bzo0NGM3MGM2OTI1MjhjMTVmOTk3OTY0NTIxYjJmOGQ2MTo3OjlmMjM6OGFjM2FhM2NiNDg1ZTRhYzMxNjY4ZjFmOGEzMzBkYWRmNzQ3NmU5MzkzOGVhZDZhYjMxMGYwYjE2NTJlMmJlNjp0OlQ6Rg. Still pretty shaky, but in case you want to see it on your own and play with it. It is essentially a 3D version of v1, and much more flexible.

Meeting Q-Ontic

(Chicago - Time Central) am9 – am8 · 2026 ,24 Feb Tuesday

Location

https://riceuniversity.zoom.us/j/5139091298?pwd=YXE0MGNKVGNEeU1Db0VUUlBNWVkrdz09
map View

Guests

organizer - Yepes Pablo

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